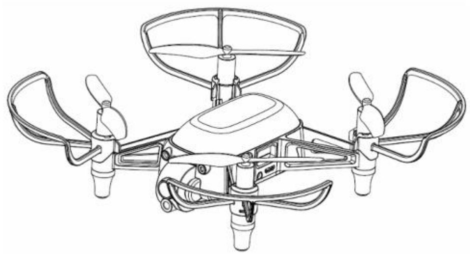

# Finals
# Game Play

Ver 3c – 14 Jul 2025

# EQUIPMENT – DRONE

Each competition team is issued with one programmable aerial drone and one computer.



- Aircraft type: Programmable quadcopter
- Aircraft wheelbase: Maximum 128mm
- Flight time per battery: 8 minutes (hovering under no wind condition, with 20% low battery protection)
- Takeoff weight: 100g
- Camera: 720P video, 1920x1080 photo resolution
- Flight stabilization: Optical flow

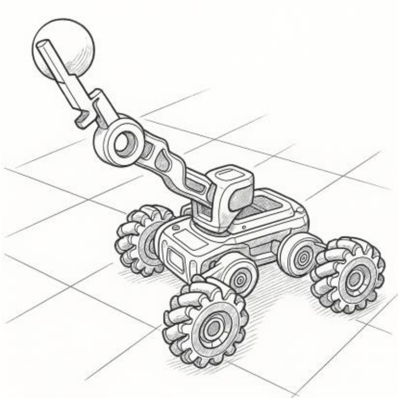
**Teams are to use the drone issued, with no modification in any way allowed.**

- Operating System: Windows 10 64bit
- CPU: Core i5 1.8Ghz 16Gb ram
- Python version: 3.6.7 (only 3.6.7 is allowed)
- IDE: Visual Studio Code
- Python libraries allowed: Opencv-python 4.6.0.60, Tflite-runtime-cp36-win64, Pyhula sdk

**While teams can use other computers, no technical support will be provided by the organisers.**

# EQUIPMENT – LAND ROBOT

Each competition team is issued with one land robot and one computer.



- Robot type: Programmable land vehicle with mechanical arm
- Robot wheelbase: 23 cm
- Operation time per battery: 3 hours under moderate use
- Camera: 720p RGB capture, 480p image transmission
- Wheels: Omnidirectional Mecanum wheels

**Teams are to use the robot issued, with no modification in any way allowed.**

- Operating System: Windows 10 64bit
- CPU: Core i5 1.8Ghz 16Gb ram
- Python version: 3.9.13 or higher
- IDE: Visual Studio Code
- Python libraries allowed: Opencv-python, numpy, ugot

**While teams can use other computers, no technical support will be provided by the organisers.**

# FINALS: Score Ranking

- Teams will each play a 20-minute match, consisting of two halves:
  - First Half (10 min) for drone obstacles – The Gauntlet, The Vortex, The Summit
  - Second Half (10 min) for land robot obstacle – The Battleground
- Every match will involve 2 teams and is synchronously-timed, i.e. start and end at the same time.
- The teams are in competition for the best total score, not just against the team in the other lane.
- Teams will be ranked by their total score, with the top 2 teams proceeding to a play-off style Grand Final match.

| Rank | School | Score |
|---|---|---|
| 1 | Everblue Sec | 440 |
| 2 | Westlight Sec | 430 |
| 3 | Silverpath High | 400 |
| 4 | Orion Sec | 390 |
| 5 | Flyfast Sec | 380 |
| | | |

Grand Final Match

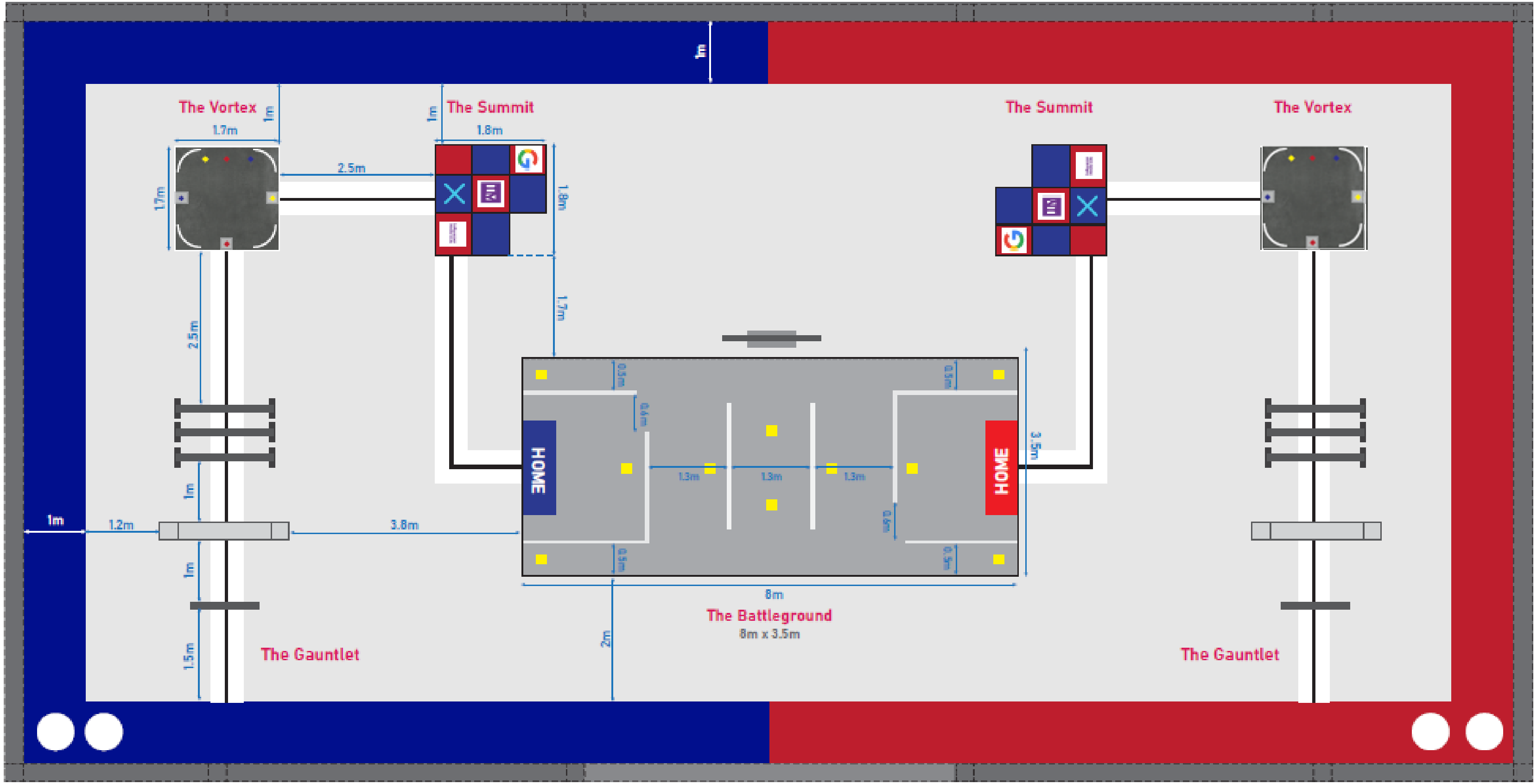
**Everblue Sec**

**VS**

**Westlight Sec**

# OVERVIEW OF OBSTACLE COURSE

- The course consists of 4 obstacles: 3 drone obstacles and 1 land robot obstacle.
- In the First Half (drone obstacles), teams can attempt the obstacles in any order and make any number of attempts with any obstacle.
- In the Second Half (land robot obstacle), teams can retrieve the point tokens in any order and make any number of attempts to do so.
- **All flight and land manoeuvre attempts must be autonomous, without any manual control or decision input.**



# OVERVIEW OF SCORING

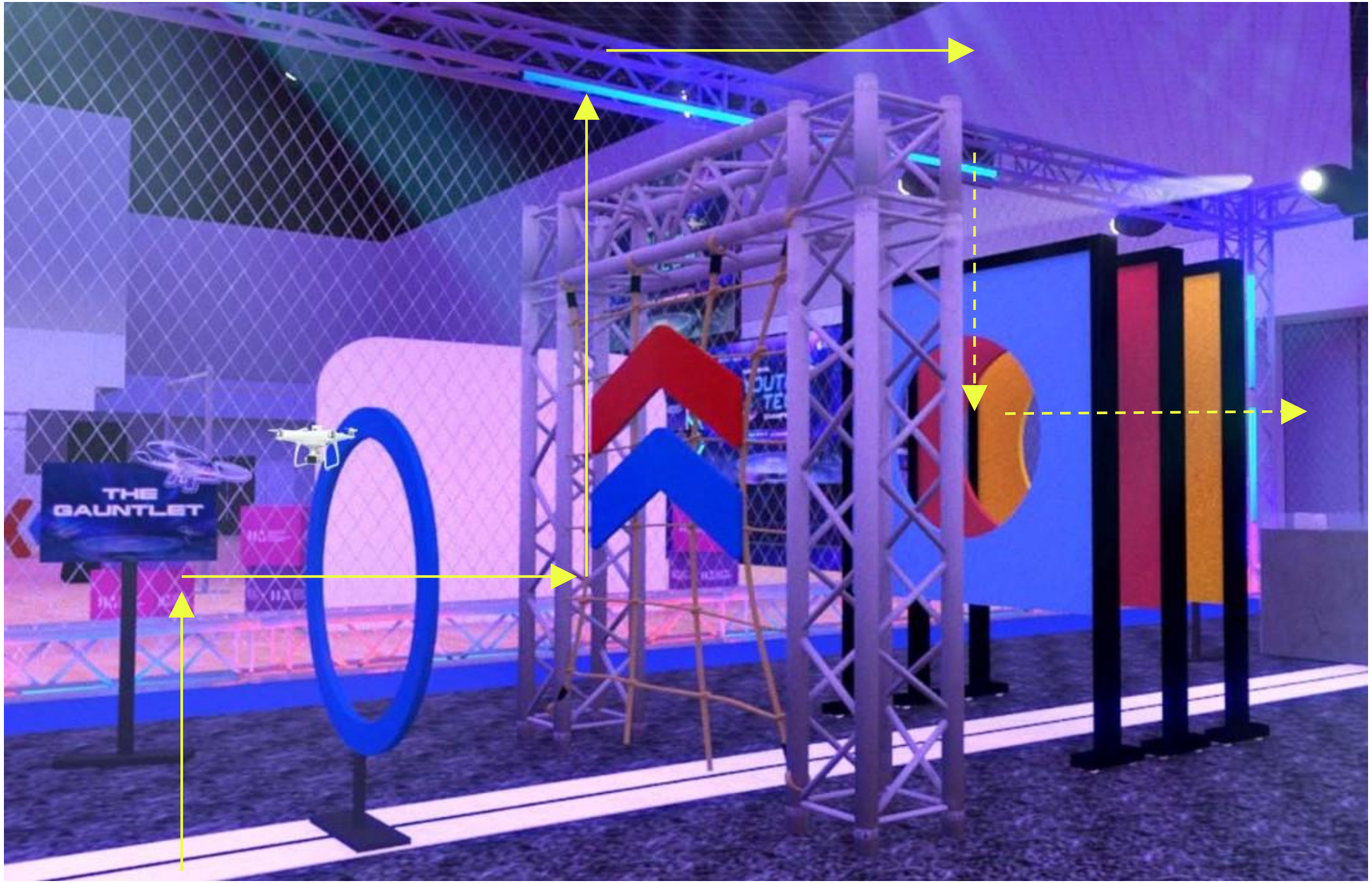
- First Half (drone obstacles): Multiple attempts allowed - the attempt with the highest points awarded will be taken as the final score for the obstacle.
- Second Half (land robot obstacle): The final score will be the sum of the point tokens successfully retrieved.
- The total score will consist of final scores from each obstacle.

| Match Period | Obstacle | Maximum Points |
|---|---|---|
| First Half (10 min) | The Gauntlet | **60** |
| | The Vortex | **165** |
| | The Summit | **135** |
| Second Half (10 min) | The Battleground | **131** |

**MAX POINTS: 491**

# THE GAUNTLET - GAMEPLAY

**Aim:** **Fly through the hoop, wall and tunnel without crashing, using sensors**



# THE GAUNTLET - SCORING & RULES

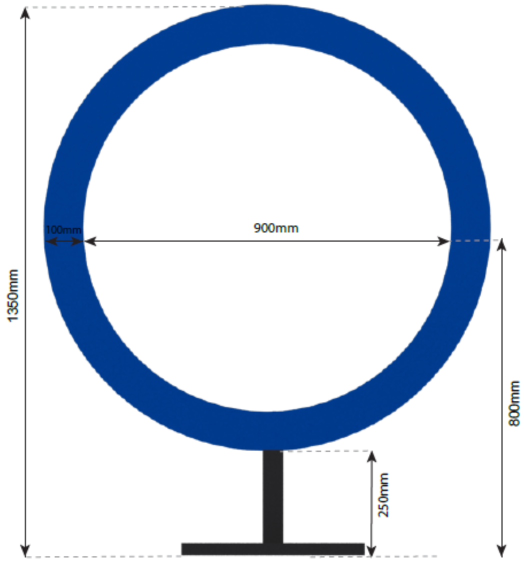
| No. | Task | Criterion | Points Awarded |
|---|---|---|---|
| 1 | Hoop | Fly through the Hoop | **10** |
| 2 | Wall | Fly over the Wall | **20** |
| 3 | Tunnel | Fly through the Tunnel | **30** |

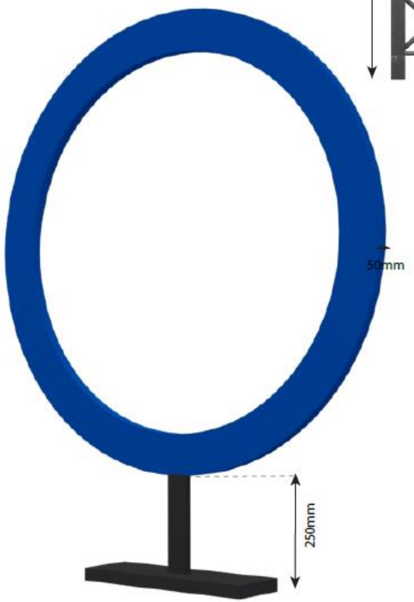
**MAX POINTS: 60**

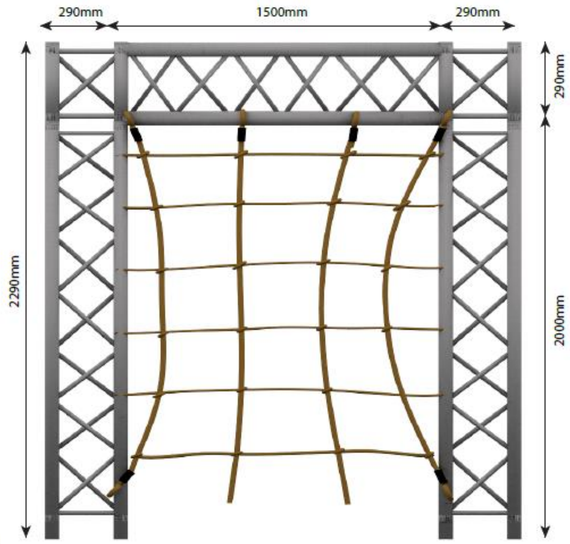
**Rules:**

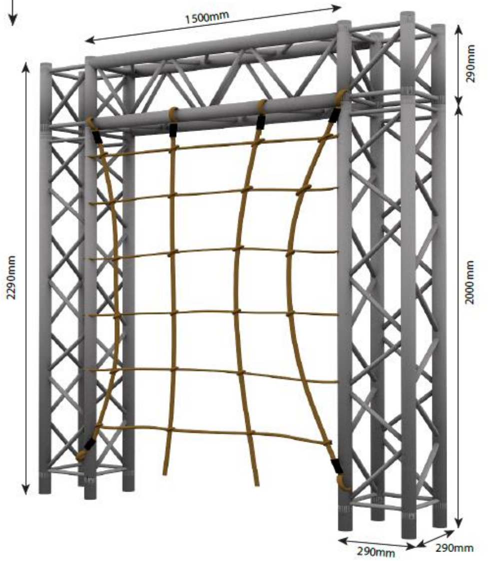
a) The drone must start or restart at the starting mission pad for all attempts.
b) An attempt ends when the flight terminates.
c) You can use the black line with white background on the ground for line tracing.
d) No other QR codes or reference signs can be added.

# THE GAUNTLET - DIMENSIONS

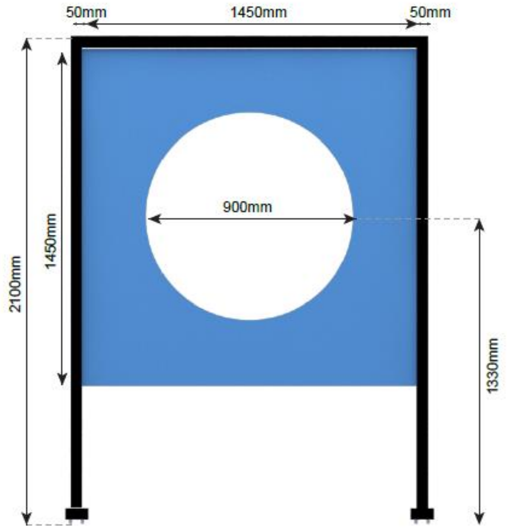


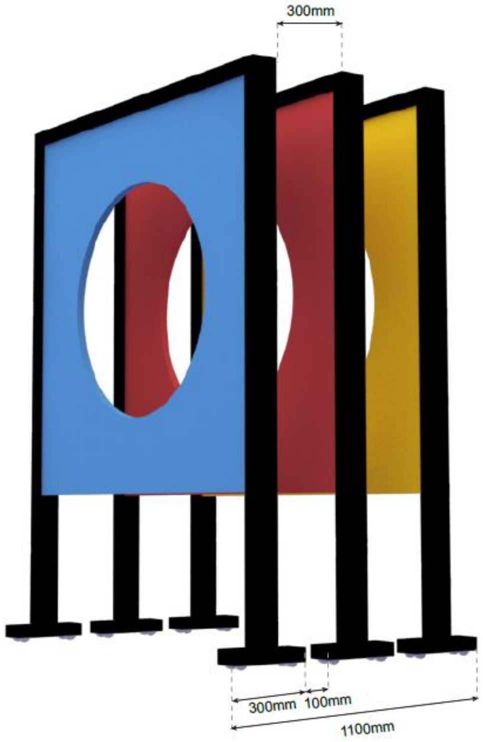



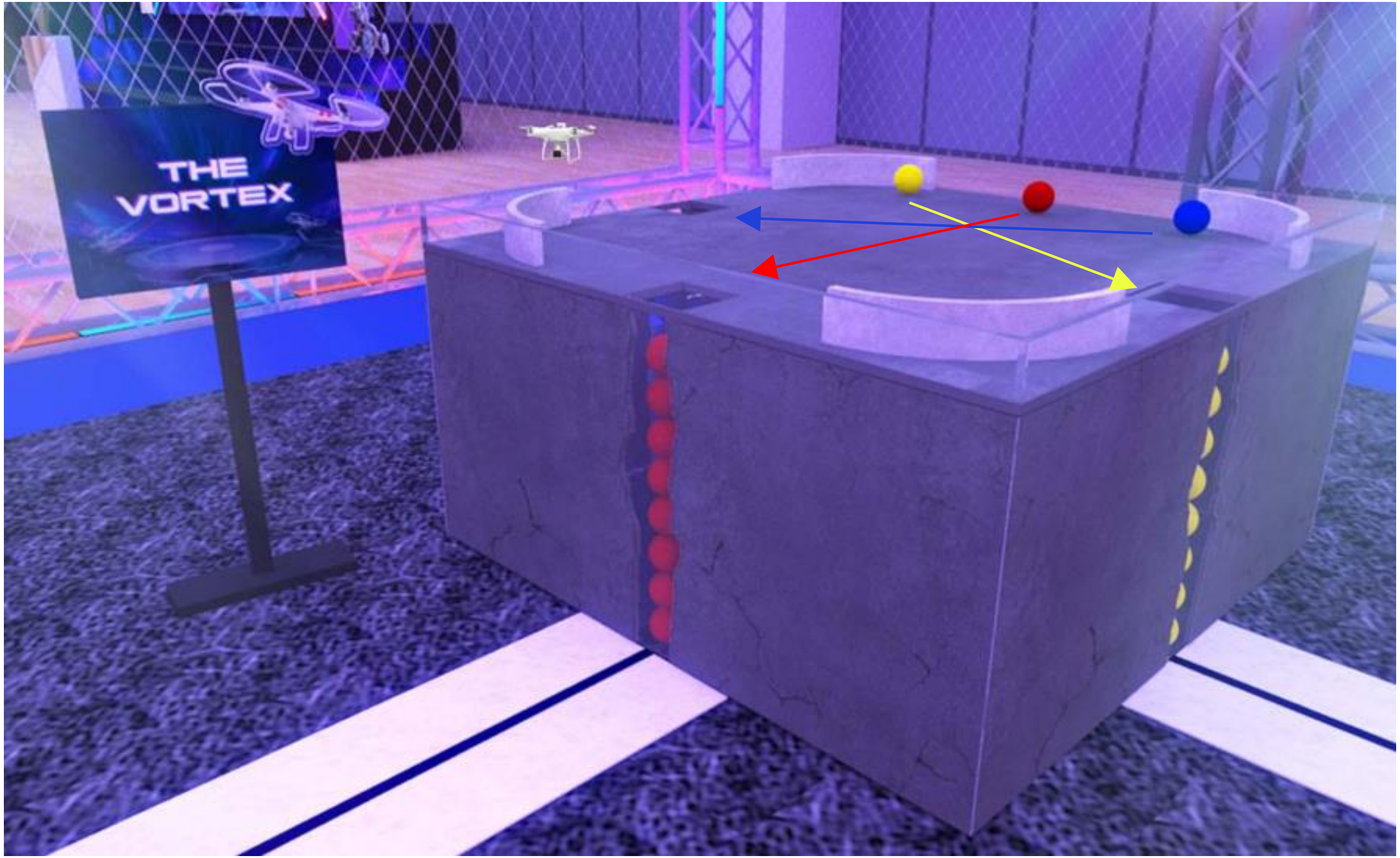
# THE GAUNTLET - DIMENSIONS



# THE VORTEX - GAMEPLAY

**Aim:** **Detect coloured balls and drive it to the correct base using propellers' downwash**



# THE VORTEX – SCORING & RULES

| No. | Task | Criterion | Points Awarded |
|---|---|---|---|
| 1 | Detect yellow ball | Show shape & colour detection of yellow ball to judge | **15** |
| 2 | Detect red ball | Show shape & colour detection of red ball to judge | **15** |
| 3 | Detect blue ball | Show shape & colour detection of blue ball to judge | **15** |
| 4 | Yellow ball enters yellow base | Must not have other balls at the base | **40** |
| 5 | Red ball enters red base | Must not have other balls at the base | **40** |
| 6 | Blue ball enters blue base | Must not have other balls at the base | **40** |

**MAX POINTS: 165**

**Rules:**

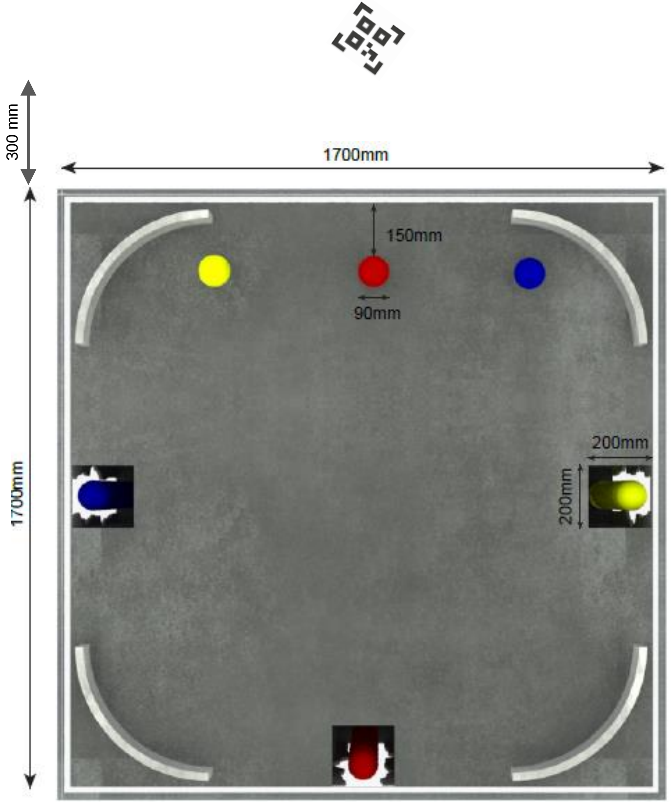
a) The drone must start or restart at the starting mission pad for all attempts.

b) An attempt ends when the flight terminates.

c) You **can add QR codes or reference signs (not provided) to guide the drone, before the attempt start.**

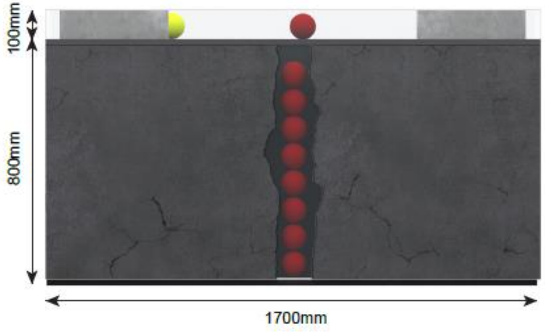
# THE VORTEX - DIMENSIONS

Top View



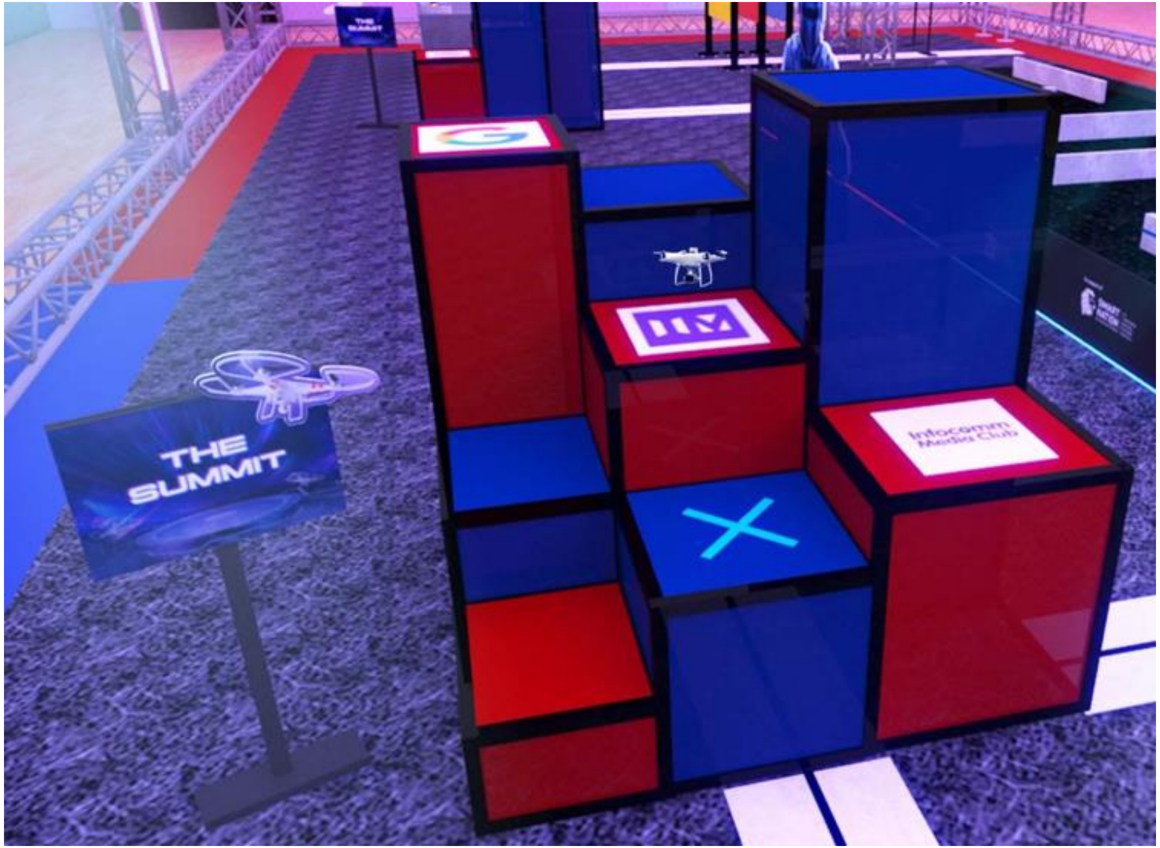

Perspective View


Front View



# THE SUMMIT - GAMEPLAY

**Aim:** **Recognise images and land on platforms of varying height and width**



# THE SUMMIT – SCORING & RULES

| No. | Task | Criterion | Points Awarded |
|---|---|---|---|
| 1 | Recognise IM logo | Show positive image recognition to judge | **15** |
| 2 | Recognise IMC logo | Show positive image recognition to judge | **15** |
| 3 | Recognise Google logo | Show positive image recognition to judge | **15** |
| 2 | Land on platforms 4, 1, 7 | Land on platform without falling off;<br>no specific order required | **25<br>per platform** |
| 3 | Land on platform 8 | Land on platform without falling off;<br>must be the last platform in the attempt | **15** |

**MAX POINTS: 135**

**Rules:**

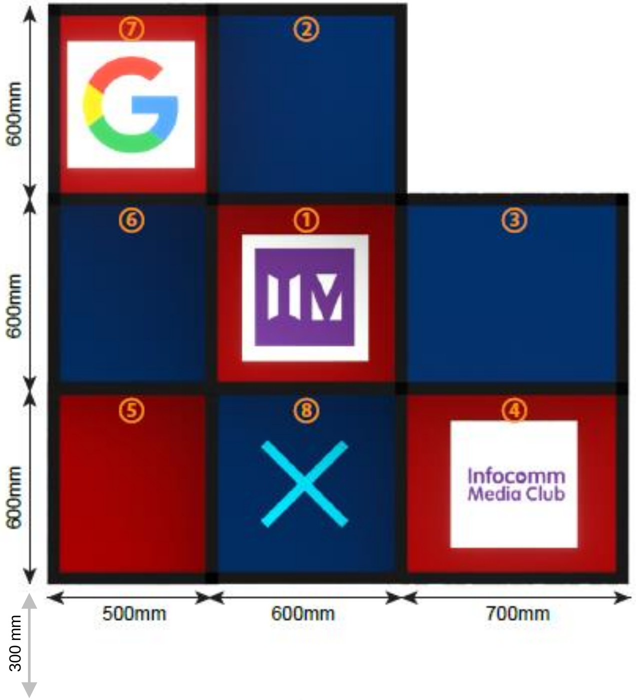
a) The drone must start or restart at the starting mission pad for all attempts.

b) An attempt ends when the drone crashes, lands on the ground, lands on platform 8, or you decide to restart.

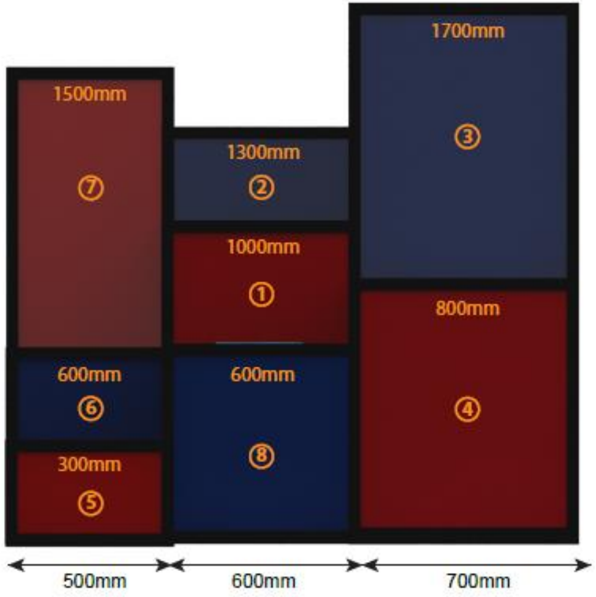
c) You **can add QR codes or reference signs (not provided) to guide the drone, before the attempt start**.

# THE SUMMIT - DIMENSIONS

Top View



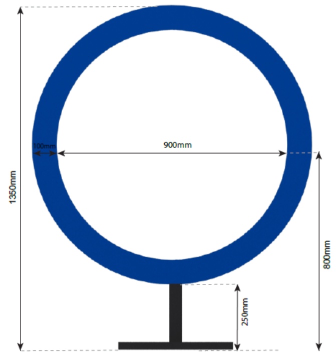

Front View

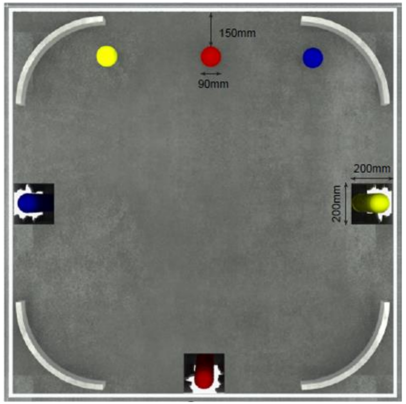


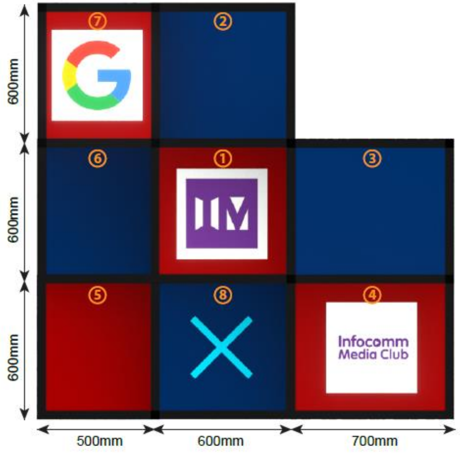

Perspective View

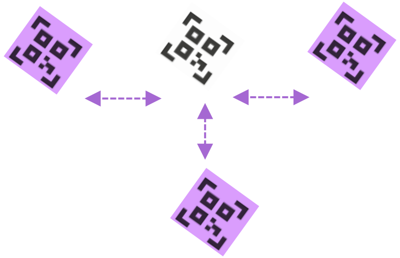
# ADJUSTING LAUNCH MISSION PADS

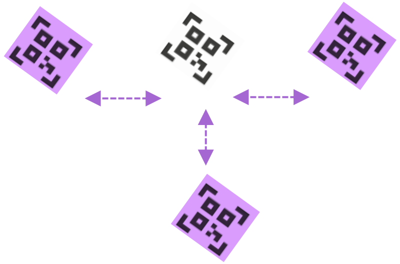
- You can adjust the Launch Mission Pads (for taking off) left or right or backwards, to compensate for drifting.
- **You cannot move the Launch Mission Pad forward (closer to the obstacle) or in midst of an attempt.**
- **You can remove the Launch Mission Pad before the attempt start, if you do need it.**

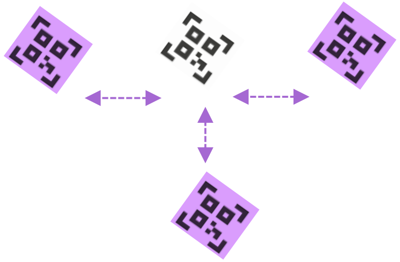






# THE BATTLEGROUND - GAMEPLAY

**Aim: Use land robot to navigate course, recognise and retrieve point tokens**

# THE BATTLEGROUND – SCORING & RULES

The final score will be the sum of the point tokens successfully retrieved (each token can be only be retrieved and scored once). Refer to the Dimensions slide for the arrangement of the point tokens.

| No. | Task | Criterion | Points Awarded |
|---|---|---|---|
| 1 | Retrieve Token 8 | Robot carries point token with arm, returns to Home area, without dropping point token | **8** |
| 2 | Retrieve Token 13 | | **13** |
| 3 | Retrieve Token 21 | | **21** |
| 4 | Retrieve Token 34 | | **34** |
| 5 | Retrieve Token 55 | | **55** |

**MAX POINTS: 131**

**Rules:**

a) The robot must start or restart at the Home area for all attempts.

**b)** **Snatching of the other team's tokens is not allowed.** The robot must keep within the permitted area of manoeuvre.

c) The robot must not intentionally pose as an obstacle blocking or interfering with the other robot, without intention to score.

d) An attempt ends when the robot returns Home with token, drops the token outside of Home, or infringes any rule. The point tokens are replaced to their original positions when an attempt ends (except for tokens retrieved to Home area).

e) No other QR codes or reference signs can be added.

# THE BATTLEGROUND - DIMENSIONS

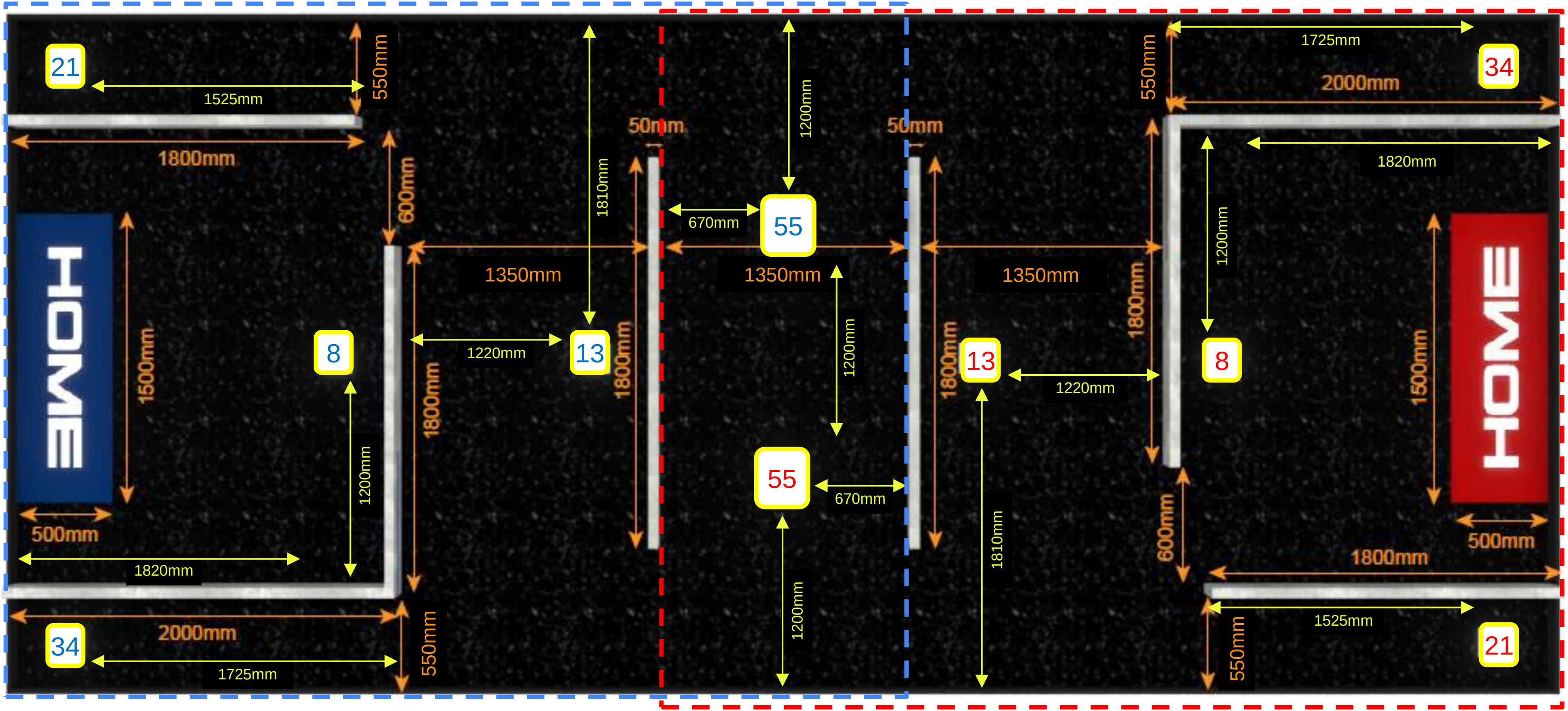
X denotes X points for Blue Team

denotes area of manoeuvre for Blue Team

Note: The team is free to decide the direction of the Point Token placement, as long as it is at the marked position.

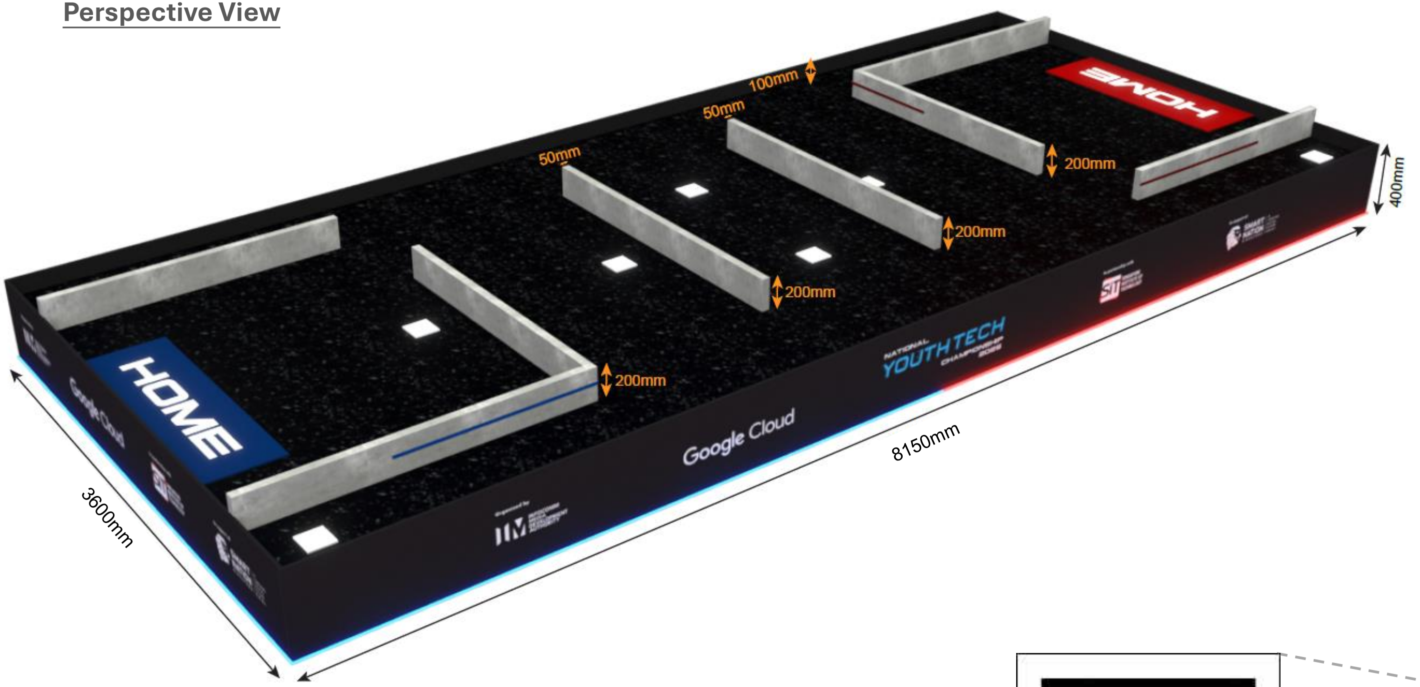
Y denotes Y points for Red Team

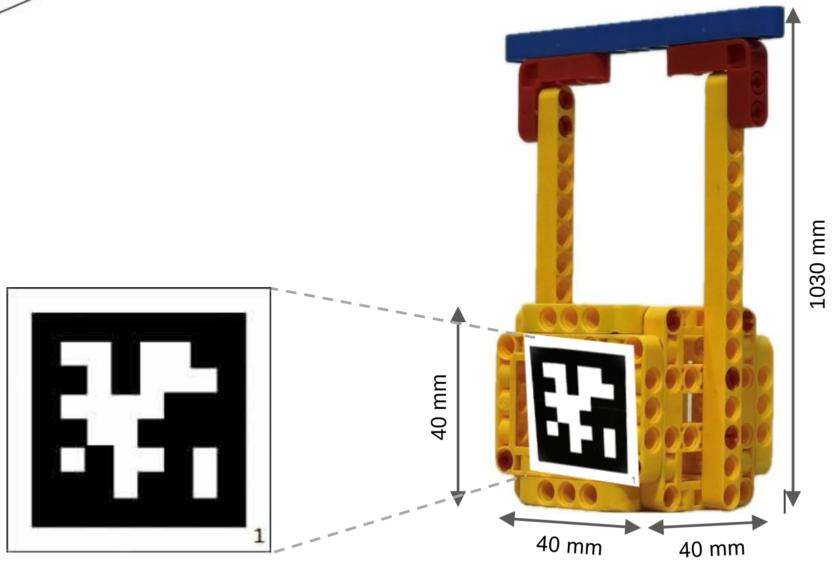
denotes area of manoeuvre for Red Team



# THE BATTLEGROUND - DIMENSIONS

## Perspective View



# FOR GRAND FINAL MATCH ONLY

- The winner of the Grand Final match will be the overall NYTC Champion.
- **Changes:** For The Battleground, **Tokens 55 and 89 are up for snatch**. The robot must keep within the permitted area of manoeuvre.

| No. | Task | Criterion | Points Awarded |
|---|---|---|---|
| 1 | Retrieve Token 8 | Robot carries point token with arm, returns to Home area, without dropping point token | **8** |
| 2 | Retrieve Token 13 | | **13** |
| 3 | Retrieve Token 21 | | **21** |
| 4 | Retrieve Token 34 | | **34** |
| 5 | **Retrieve Token 55 (up for snatch by other team)** | | **55** |
| 6 | **Retrieve Token 89 (up for snatch by other team)** | | **89** |

MAX POINTS: 220

- **Changes:** The **Battleground will be worth 220 points** in the total score.

| Match Period | Obstacle | Maximum Points |
|---|---|---|
| First Half (10 min) | The Gauntlet | **60** |
| | The Vortex | **165** |
| | The Summit | **135** |
| Second Half (10 min) | **The Battleground** | **220** |

MAX POINTS: 580

# FOR GRAND FINAL MATCH ONLY: THE BATTLEGROUND - DIMENSIONS

X denotes X points for Blue Team

denotes area of manoeuvre for Blue Team

X Y denotes X points for Blue Team or Y points for Red Team, whichever retrieves it successfully

denotes direction where apriltag faces

Y denotes Y points for Red Team

denotes area of manoeuvre for Red Team